

REVIEW

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Beyond access: WASH, public health, and sustainable development priorities in Africa—a narrative review

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Abstract

Background Water, sanitation, and hygiene [WASH] are core determinants of public health, health equity, and sustainable development in Africa. Despite progress in service coverage, major gaps persist in safety, reliability, affordability, quality, and sustainability. However, evidence remains fragmented on how these challenges collectively affect public health and sustainable development across African settings.

Aim This narrative review synthesizes current evidence to address fragmented knowledge on how these challenges influence public health and sustainable development across African settings.

Methods A narrative review was conducted using peer-reviewed articles, reviews, global health reports, policy documents, and technical resources identified through PubMed, Scopus, Web of Science, Google Scholar, WHO, UNICEF, UN-Water, and WaterAid. Evidence was synthesized thematically across five domains: WASH inequalities, public health burden, social and system-level barriers, governance and infrastructure sustainability, and evidence-based priorities.

Results WASH inequalities extend beyond service coverage to include deficiencies in safety, reliability, affordability, and sustainability. Unsafe water, inadequate sanitation, and poor hygiene remain major contributors to preventable disease and health-system vulnerability. Persistent social, behavioral, and system-level barriers, along with governance, financing, infrastructure maintenance, and climate vulnerability, hinder sustainable improvements in WASH. Strengthening governance, resilient infrastructure, community participation, and multisectoral collaboration emerged as key priorities for improving public health and sustainable development.

Conclusions Africa's WASH agenda must move beyond access toward safe, reliable, equitable, climate-resilient, and sustainable services. Integrated WASH action can reduce preventable disease, protect vulnerable populations, strengthen health-system resilience, and accelerate progress toward the Sustainable Development Goals.

Keywords Water, Sanitation, Hygiene, Public health, Health equity, Sustainable development, Africa

Introduction

Water, sanitation, and hygiene [WASH] are fundamental determinants of public health, human dignity, health equity, and sustainable development [1, 2]. In Africa, WASH remains a persistent public health and development challenge despite global and national commitments to expand access to safe drinking water, sanitation, and hygiene services [1, 2]. The UNICEF and WHO report on household drinking water, sanitation, and hygiene from

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2000 to 2017 documented progress in service coverage yet confirmed substantial inequalities across income groups, geographic settings, and vulnerable populations [1]. Recent evidence further demonstrates that WASH practices in Africa directly influence public health outcomes and sustainable development, particularly in communities affected by poverty, weak infrastructure, displacement, and limited health resources [2]. Accordingly, WASH is a central determinant of disease prevention, health equity, and long-term development, rather than merely a basic service issue.

Barriers to WASH improvement in Africa are multi-dimensional and interconnected. A scoping review from Southern Africa identified social, economic, environmental, and institutional factors influencing WASH behavior and service use, including household income, water availability, knowledge, cultural practices, and infrastructure reliability [3]. These barriers produce direct public health consequences. The World Health Organization recognizes mortality attributed to unsafe WASH services as an indicator under Sustainable Development Goal 3.9.2, confirming the role of unsafe water, inadequate sanitation, and poor hygiene in preventable deaths [4]. In sub-Saharan Africa, progress in sanitation is constrained by weak systems, limited investment, and gaps between policy targets and implementation [5]. WASH challenges, therefore, reflect broader failures in governance, financing, infrastructure, and equity.

Sustainable WASH improvement requires strong systems, reliable monitoring, and accountable governance. The WHO/UN-Water Global Analysis and Assessment of Sanitation and Drinking Water emphasized the roles of national systems, financing, human resources, planning, and accountability in improving sanitation and drinking water outcomes [6]. Systems-based approaches further show that durable WASH progress depends on interactions among infrastructure, institutions, community behavior, environmental conditions, and service delivery mechanisms [7]. Health education also contributes to WASH improvement. Evidence from Nigeria showed that targeted health education improved water, sanitation, and hygiene practices among adolescent girls, demonstrating the value of sustained community-level behavior-change interventions [8].

Unsafe WASH continues to impose a substantial public health burden. The WHO Global Water, Sanitation, and Hygiene Annual Report 2022 called for stronger action to close WASH gaps and protect vulnerable populations [9]. Earlier global burden studies demonstrated that inadequate water, sanitation, and hygiene contribute substantially to disease burden in low- and middle-income countries [10]. Updated analyses confirmed the contribution of unsafe WASH to adverse health outcomes,

particularly where poverty, environmental exposure, and weak service systems intersect [11]. The WHO burden of disease assessment also identifies unsafe drinking water, sanitation, and hygiene as preventable causes of illness and death [12]. Recent global evidence from the Lancet further confirms that unsafe WASH in domestic settings contributes to avoidable adverse health outcomes, with low-resource settings carrying a disproportionate burden [4, 13].

The effects of WASH extend beyond infectious disease. WASH conditions shape education, gender equity, economic productivity, environmental health, and resilience to public health emergencies. Hygiene improvement in developing countries is constrained by poverty, low awareness, poor infrastructure, weak policy enforcement, and limited sustainability of behavior change [14]. Country-level experiences, including Nigeria, show that achieving Sustainable Development Goal 6 requires stronger policy implementation, financing, and accountability [1, 9, 15]. Water infrastructure sustainability remains a critical concern because service systems fail when maintenance, governance, technical capacity, and long-term financing are weak [2, 16]. These challenges are highly relevant across Africa, where WASH gaps undermine health systems and delay progress toward sustainable development.

Although previous reviews, including Okesanya et al. (2024), have examined WASH practices and service coverage in Africa, evidence remains fragmented regarding how WASH inequalities, governance, infrastructure sustainability, climate resilience, and health equity collectively influence public health and sustainable development [2]. This review addresses this gap by providing an integrated synthesis of these interconnected dimensions to inform policy, practice, and future research. This narrative review synthesizes evidence on WASH, public health, and sustainable development priorities in Africa. The review is important because it moves the discussion beyond access and reframes WASH as an integrated priority in public health, equity, governance, infrastructure, and sustainability. By consolidating evidence on WASH inequalities, disease burden, system-level barriers, governance, infrastructure sustainability, and development priorities, this review provides a foundation for strengthening safe, reliable, equitable, climate-resilient, and sustainable WASH services across African settings.

Methods

Review design

This narrative review was undertaken to synthesize and critically examine evidence on water, sanitation, and hygiene [WASH] in Africa and its implications for public health, health equity, governance, infrastructure

sustainability, and sustainable development. A narrative review approach was selected because the available literature spans multiple disciplines, including public health, environmental health, health systems, policy, climate resilience, and development studies, and includes diverse evidence sources such as empirical studies, systematic reviews, policy reports, and technical documents. This approach enabled the integration of evidence from heterogeneous sources and facilitated a comprehensive examination of major themes, challenges, and priorities relevant to WASH across African settings [17, 18].

Conceptual framework

The conceptual framework (Fig. 1) was developed to guide the review by illustrating the relationships among WASH inputs, governance, financing, community participation, health education, and climate resilience, and their influence on public health, health equity, and sustainable development in Africa. The framework informed the literature search by identifying key review domains, guided the organization and thematic synthesis of the retrieved evidence, and provided the structure for presenting the review findings.

Search strategy

This narrative review was conducted to synthesize current evidence on water, sanitation, and hygiene [WASH]

practices in Africa and their implications for public health, health equity, governance, infrastructure sustainability, and sustainable development. A structured literature search was conducted in PubMed, Scopus, Web of Science, Google Scholar, and the WHO Institutional Repository for Information Sharing [IRIS]. The search included publications published from 2000 to March 2026, with a primary focus on studies conducted in Africa. Global evidence was included where it provided relevant conceptual, methodological, or policy insights applicable to the African context.

The search terms included combinations of “water, sanitation, and hygiene,” “WASH,” “Africa,” “sub-Saharan Africa,” “public health,” “safe drinking water,” “sanitation,” “hygiene practices,” “cholera,” “diarrheal disease,” “health equity,” “WASH in health facilities,” “WASH in schools,” “menstrual hygiene,” “climate change,” “water security,” “infrastructure sustainability,” “community participation,” and “sustainable development goals.” Boolean operators such as AND and OR were used to combine related terms. The reference lists of relevant articles and reports were also screened to identify additional sources.

To strengthen the policy relevance of the review, grey literature from authoritative organizations was also considered. These included reports, technical documents, and policy materials from the World Health Organization, UNICEF, UN-Water, WaterAid, and other

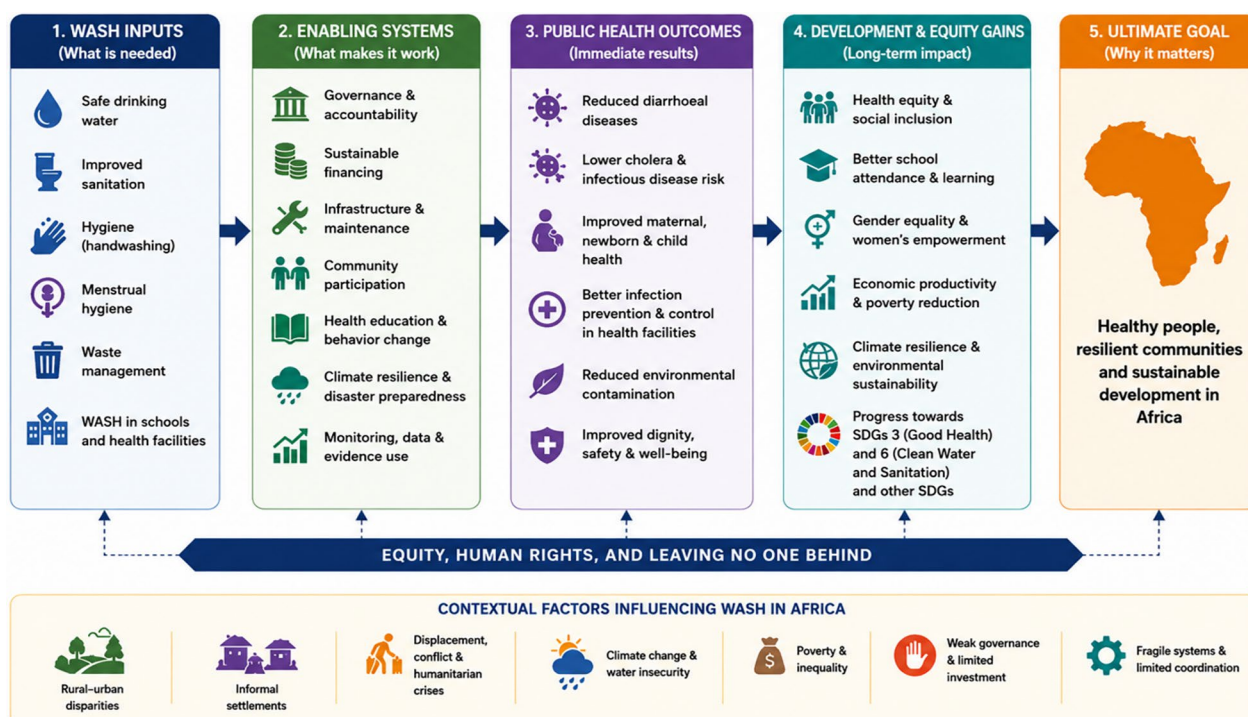


Fig. 1 Conceptual framework of WASH and sustainable development in Africa

recognized public health and development organizations. These sources were included when they provided important evidence, guidance, or contextual information on WASH systems, health risks, equity, and sustainable development.

Eligibility criteria

This review included peer-reviewed original research articles, systematic reviews, scoping reviews, narrative reviews, policy briefs, global health reports, and technical documents that addressed WASH access, practices, health outcomes, governance, sustainability, climate vulnerability, health facilities, schools, humanitarian settings, or sustainable development in Africa. Sources were considered eligible when they provided evidence relevant to WASH-related disease burden, equity gaps, behavioral and system-level barriers, infrastructure sustainability, or policy and programmatic responses.

Only English-language publications were included. However, documents with English abstracts or summaries were considered when they were directly relevant to the review topic. Evidence from African countries was prioritized, but global and low- and middle-income country evidence was included when it offered important conceptual, methodological, or policy relevance to the African WASH context. Grey literature was included only when it originated from authoritative organizations, demonstrated clear institutional authorship, and provided relevant, evidence-based guidance or data on WASH, public health, governance, or sustainable development.

Sources were excluded if they were opinion pieces without supporting evidence, non-authoritative blogs, duplicate publications, documents with limited relevance to WASH or public health, or reports lacking sufficient methodological or institutional credibility. Publications focused solely on engineering design and lacking clear relevance to public health, equity, governance, or sustainable development were also excluded.

Study selection and data extraction

Retrieved records were screened for relevance based on titles, abstracts, and, when necessary, full texts. Duplicate publications were removed during the screening process. Sources meeting the eligibility criteria were retained for inclusion. Information from eligible sources was extracted and organized using a structured evidence summary form. Extracted information included publication type, geographical setting, study focus, key findings, and relevance to WASH, public health, governance, sustainability, or development outcomes. A total of 60 sources were included in the final review.

Evidence synthesis

Given the diversity of study designs, populations, settings, outcomes, and policy contexts represented in the literature, a thematic synthesis approach was employed to identify, organize, and interpret recurring patterns across the included evidence, consistent with established qualitative synthesis methodology [19]. Findings were compared across studies and synthesized into five thematic domains: WASH inequalities, disease burden, social and system-level barriers, governance and infrastructure sustainability, and strategic priorities for strengthening WASH within the public health and sustainable development agenda. As this was a narrative review, a formal quality appraisal was not undertaken. Instead, the interpretation of the evidence considered the methodological rigor, consistency of findings, and relevance of the included sources.

A total of 60 sources were included in the review. These consisted of peer-reviewed research articles, systematic and scoping reviews, global health reports, policy documents, and technical resources from international organizations. This approach allowed the review to combine academic evidence with policy-relevant insights and to present WASH as a cross-cutting priority for public health, equity, resilience, and sustainable development in Africa.

Results and discussion

WASH inequalities and the limits of access in Africa

The findings indicate that WASH inequalities in Africa are driven not only by limited-service coverage but also by deficiencies in service safety, reliability, affordability, accessibility, and long-term sustainability. Effective access depends on service safety, reliability, affordability, physical reach, cultural acceptability, and long-term sustainability. In many African settings, WASH services are shaped by political, economic, agricultural, environmental, and institutional decisions. Multi-stakeholder approaches are essential because water management affects households, agriculture, health systems, local authorities, and national development priorities [6, 20]. This is particularly important in rural Africa, where water systems remain underfinanced, and communities often bear the long-term costs of service maintenance without adequate technical or financial support [6, 21] (Fig. 2).

WASH inequities are closely linked with broader health and development disparities. Evidence from Nigeria shows that investment in health and social systems is essential for improving population well-being and reducing structural inequities [22]. Poor water and sanitation conditions increase the risk of disease, reduce dignity, limit productivity, and weaken household resilience.

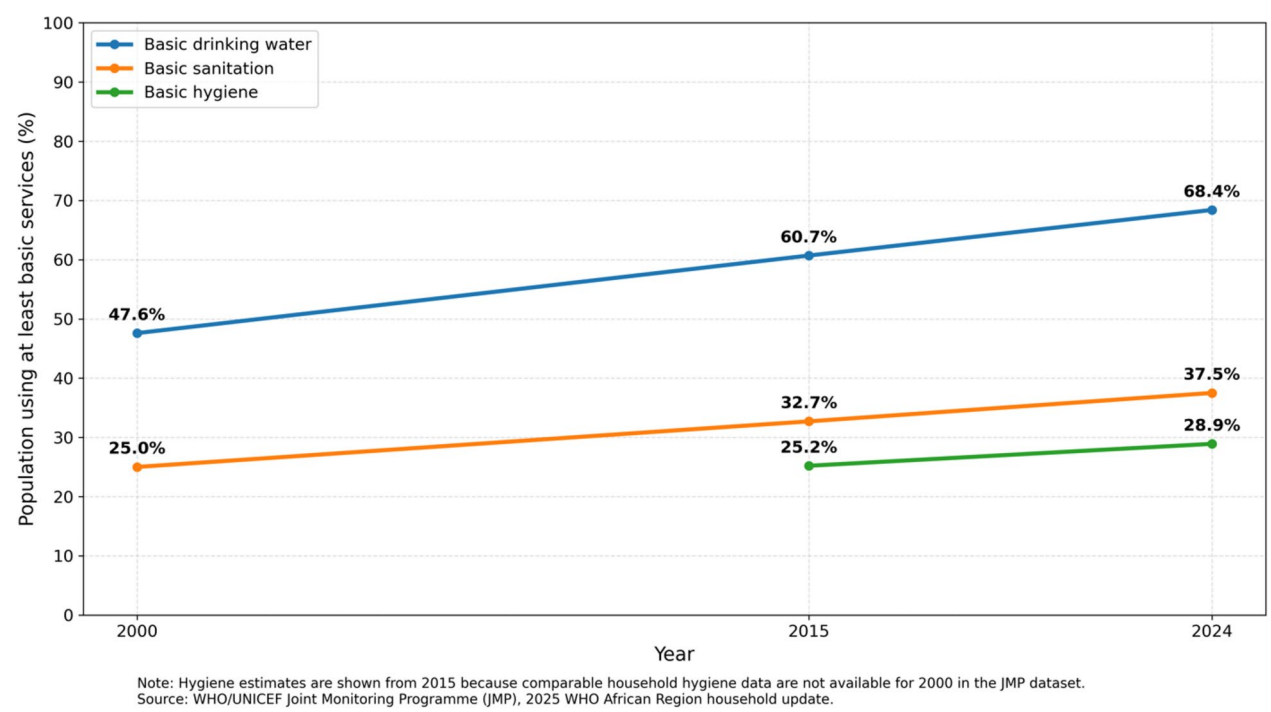


Fig. 2 Trends and inequalities in access to drinking water, sanitation, and hygiene services in Africa

Climate change intensifies these inequalities through water scarcity, flooding, environmental instability, and increased risk of cholera and other WASH-related diseases [14, 23]. In several African countries, uneven climate-related health preparedness makes WASH resilience an urgent public health priority [7, 20, 24] (Fig. 3).

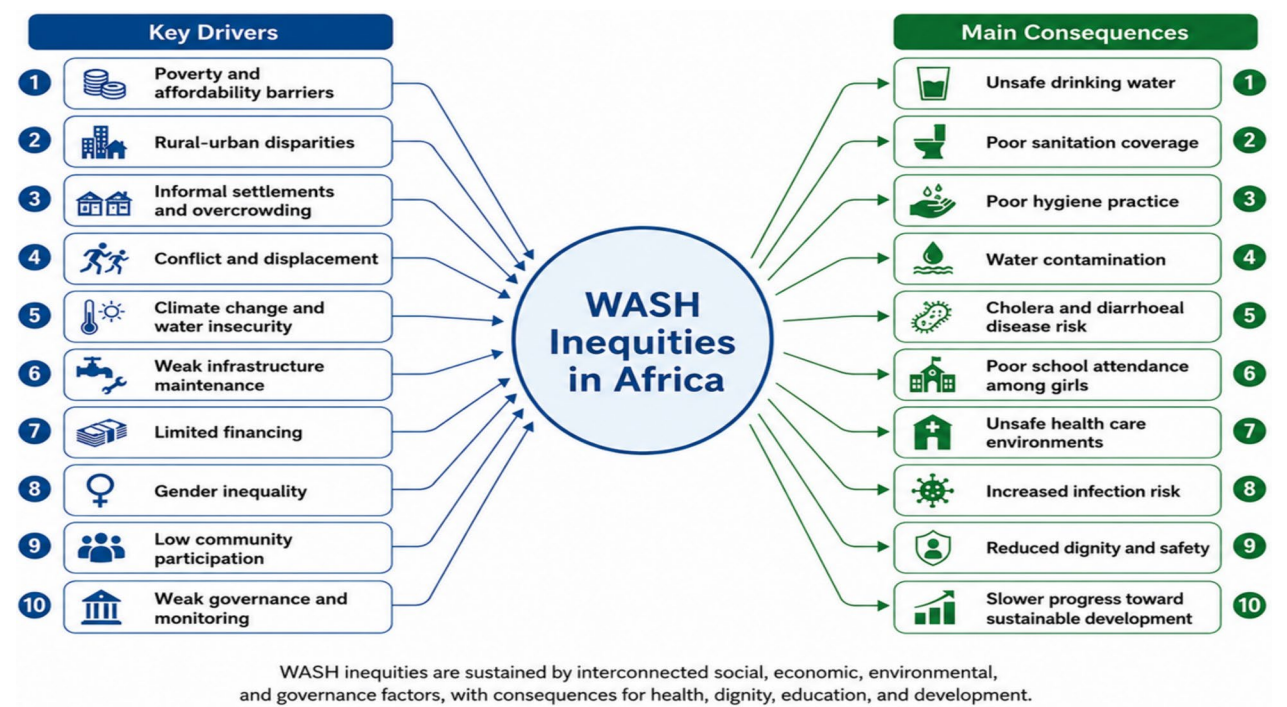


Fig. 3 Drivers and consequences of WASH inequities in Africa

These inequities are sustained by interconnected social, economic, environmental, and governance determinants, including poverty, rural–urban disparities, informal settlements, conflict, displacement, climate change, weak infrastructure maintenance, limited financing, gender inequality, low community participation, and weak monitoring systems [6, 24].

The interaction between climate change, sanitation systems, and groundwater quality further illustrates the complexity of WASH challenges in Africa. Evidence from Botswana demonstrates how environmental change can compromise the safety and sustainability of domestic water sources by affecting sanitation infrastructure and groundwater quality [25]. These findings show that WASH performance should be evaluated not only by infrastructure availability but also by service safety, reliability, environmental resilience, and long-term functionality.

WASH inequalities are also evident in humanitarian and displacement settings. Evidence from refugee camps and settlements in Bangladesh, Kenya, Uganda, South Sudan, and Zimbabwe provides transferable insights into social and geographic inequalities in access to water, sanitation, and hygiene services [26]. Although some of this evidence originates from non-African settings, it provides valuable contextual insights for humanitarian WASH responses in Africa and should be interpreted in light of the respective contextual differences. In these settings, poor WASH conditions increase the risk of disease and undermine dignity, safety, and resilience.

Collectively, the evidence indicates that WASH inequalities are rooted in broader social, environmental, economic, and governance determinants that shape health opportunities and development outcomes. Addressing these inequalities requires integrated strategies that strengthen infrastructure, governance, climate resilience, and equity-focused public health interventions.

Public health burden of unsafe water, sanitation, and hygiene

Unsafe water, poor sanitation, and inadequate hygiene remain major drivers of preventable disease across Africa. Evidence from peri-urban Lusaka, Zambia, showed that targeted behavior-change interventions improved sanitation outcomes, confirming the value of community-level strategies in reducing WASH-related risks [26]. However, behavior change must be supported by safe environments, a reliable water supply, adequate sanitation infrastructure, and reduced overcrowding. In informal settlements, such as Goreangab in Windhoek, Namibia, limited access to safe drinking water illustrates how urban poverty and inadequate planning expose vulnerable communities to unsafe WASH conditions [27].

Water contamination is a direct pathway linking inadequate WASH to adverse health outcomes. Evidence from rural villages in Lesotho identified bacterial contamination of drinking water sources associated with neighborhood sanitation and hygiene practices [28]. Such contamination increases exposure to diarrheal diseases and other infections, particularly among children and households with limited access to health services. Cholera resurgence in Africa further demonstrates the public health consequences of weak WASH systems. Recurrent outbreaks expose persistent gaps in surveillance, preparedness, water safety, sanitation coverage, and rapid public health response despite global elimination targets [13, 29].

WASH must be treated as a core component of disease prevention and outbreak control. Global and public health organizations emphasize that water, sanitation, and hygiene are essential for protecting health and preventing the transmission of infectious diseases [30, 31]. In sub-Saharan Africa, the combined effect of unsafe water and poor sanitation practices has been associated with childhood diarrhea, demonstrating that WASH risks operate together rather than separately [22, 32]. Interventions that address only one component, such as water supply without sanitation or hygiene, produce a limited public health impact. A comprehensive approach that integrates safe water, sanitation, hygiene behavior, surveillance, and community engagement is required to reduce preventable disease and strengthen outbreak resilience.

Social, behavioral, and system-level barriers to sustainable WASH practices

Sustainable WASH practices are shaped by social, behavioral, gender-related, and institutional determinants. UNICEF identifies WASH as essential for child health, dignity, education, and survival, particularly in vulnerable communities [1, 22, 33]. Daily barriers continue to limit safe hygiene practices, including inadequate facilities, limited privacy, insufficient materials, and social constraints. Menstrual hygiene management illustrates this challenge. Among adolescent schoolgirls in Taraba State, Nigeria, menstrual hygiene challenges were linked to knowledge, facilities, privacy, and access to materials [34]. These barriers affect school participation, dignity, and gender equity, demonstrating the close relationship between WASH and social inclusion [35].

Water security is closely linked to economic growth, national development, and environmental health. Evidence on water security and sustainable growth emphasizes that secure water systems are essential for development, productivity, and resilience [13, 24, 36]. Climate variability and change increase health risks and

economic costs, particularly in countries with limited adaptive capacity [37]. Environmental sanitation is further affected by waste management practices. Poor solid waste management in the Global South has major implications for environmental sustainability [38], while open dumping and landfilling contribute to environmental pollution and health risks [39]. These findings demonstrate that sustainable WASH requires integrated action across safe water, sanitation, hygiene, waste management, climate adaptation, and community-level behavior change.

WASH systems, governance, and infrastructure sustainability

Strong WASH systems require governance, financing, trained personnel, accountability, and long-term maintenance. In health facilities across sub-Saharan Africa, uneven access to WASH services has direct implications for infection prevention and control, including during outbreaks such as COVID-19 [16, 40]. Health facilities without safe water, toilets, hand hygiene, waste management, and environmental cleaning cannot provide safe care. In conflict settings, WASH interventions for women and children become more urgent because displacement and insecurity increase vulnerability to disease [24, 31, 41].

WASH in health care settings is also central to reducing antimicrobial resistance. Poor water, sanitation, and hygiene increase infection transmission, antibiotic use, and the risk of resistant infections [42]. In traditional birth homes in Southwest Nigeria, gaps in WASH facilities and infection prevention practices expose mothers, newborns, birth attendants, and communities to preventable risks when WASH is not integrated into maternal and child health services [43]. Menstrual hygiene challenges in rural schools of Zambia further show how weak WASH infrastructure affects girls' education, dignity, and participation [34, 44]. These examples demonstrate that WASH systems must extend across households, schools, health facilities, and community institutions.

WASH sustainability is closely linked to the broader development agenda. Environmental sustainability and strong government institutions are essential for achieving development goals [4, 22, 26, 45]. Investment in WASH generates economic benefits by reducing disease, improving productivity, supporting education, and strengthening resilience [11, 13, 46]. A holistic approach to water, sanitation, and public health connects infrastructure with behavior change, governance, community engagement, and environmental protection [47]. Rural public health can be strengthened through best-practice WASH initiatives adapted to local needs and supported by long-term systems [44, 48]. Capacity building remains essential; community-led total sanitation management training in Kenya demonstrates the importance of developing local

leadership and implementation skills for sustainable WASH programming [49].

Evidence-based priorities for strengthening WASH as a public health and sustainable development agenda

Strengthening WASH as a public health and sustainable development priority requires approaches that extend beyond infrastructure provision. Sustainable WASH interventions must align with community needs, social realities, and contextual constraints, as behavior-change initiatives are more effective when they reflect local motivations, capacities, and priorities [34, 50]. In health facilities, tools such as WASH-FIT support the assessment of WASH conditions and facilitate the linking of facility readiness, patient satisfaction, and maternal health outcomes. Evidence from Indonesia demonstrates the potential value of this approach and provides transferable implementation lessons that may inform adaptation in African healthcare settings [51]. Lessons from primary care facilities during the COVID-19 pandemic further demonstrate that WASH improvements are fundamental to safe care, outbreak preparedness, and health-system resilience [36, 52].

Achieving sustainable progress also requires stronger governance, accountability, and long-term financing. The global WASH crisis highlights the need for coordinated action across sectors and institutions [53]. National and organizational strategies, including WaterAid Nigeria's 2023–2028 program strategy, emphasize the importance of integrating water, sanitation, hygiene, equity, and institutional strengthening into long-term planning frameworks [1, 15, 54]. Policy lessons from the post-2015 development agenda similarly underscore the need to maintain WASH as a priority within national and regional development plans to achieve universal and equitable services [14, 55]. Evidence addressing Sustainable Development Goal targets further confirms that progress depends on evidence-informed planning, effective monitoring systems, sustained financing, and implementation capacity [26, 56].

Community participation remains central to sustainable WASH systems. Meaningful community involvement in planning, implementation, and evaluation can strengthen ownership, accountability, service sustainability, and responsiveness to local needs [57]. Evaluation approaches that capture meaningful community-level change provide valuable insights into how WASH interventions influence health, well-being, and quality of life beyond infrastructure outputs alone [58]. Evidence from a realist review further demonstrates that community participation contributes to improved health outcomes, infrastructure sustainability, and service longevity in low- and middle-income countries [59]. Sustained adoption of

Table 1 Key WASH challenges in Africa and recommended public health and sustainable development responses

WASH challenge	Public health and development implications	Recommended response
Unequal access to safe water and sanitation	Increases health inequities, especially in rural communities, informal settlements, and displaced populations	Prioritize equity-focused WASH planning, targeted financing, and service expansion for underserved groups
Unsafe drinking water and poor sanitation	Contributes to diarrhoeal diseases, cholera, parasitic infections, and environmental contamination	Strengthen water safety, sanitation systems, hygiene promotion, surveillance, and outbreak preparedness
Weak WASH infrastructure maintenance	Causes service breakdown, unreliable access, and loss of community trust	Establish long-term maintenance systems, local technical capacity, spare-part supply chains, and accountability mechanisms
Limited hygiene behavior change support	Reduces sustained handwashing, safe water storage, latrine use, and menstrual hygiene practice	Integrate community-based health education with access to water, soap, sanitation facilities, and culturally acceptable services
Poor WASH in health facilities	Increases infection risk, weakens infection prevention and control, and threatens maternal and newborn safety	Ensure safe water, toilets, hand hygiene stations, waste management, and environmental cleaning in all health facilities
Poor WASH in schools	Affects school attendance, menstrual hygiene, dignity, and learning outcomes, especially among girls	Provide gender-sensitive toilets, handwashing facilities, menstrual hygiene materials, and school-based WASH education
Climate change and water insecurity	Increases drought, flooding, contamination, cholera risk, and service disruption	Build climate-resilient WASH systems, protect groundwater, improve emergency preparedness, and integrate WASH into climate adaptation plans
Weak governance and fragmented coordination	Limits sustainability, financing, monitoring, and accountability	Strengthen multi-sector coordination between health, water, education, environment, urban planning, and emergency response sectors
Limited community participation	Reduces ownership, acceptability, and long-term sustainability of WASH interventions	Involve communities in planning, implementation, monitoring, and evaluation of WASH programmes
Narrow measurement of WASH progress by access alone	Misses service quality, safety, affordability, reliability, and dignity	Measure WASH progress using quality, equity, functionality, sustainability, and health-impact indicators

safe water, hygiene, and sanitation technologies depends on social acceptance, affordability, maintenance capacity, trust, and long-term support mechanisms [60]. Collectively, the evidence indicates that successful WASH programs should be evaluated not only by service coverage but also by their ability to deliver safe, equitable, reliable, sustainable, and meaningful benefits to the populations they serve (Table 1).

Strengths and limitations

This narrative review synthesizes evidence from peer-reviewed studies, systematic reviews, policy documents, and reports from international organizations to provide a comprehensive overview of priorities in water, sanitation, and hygiene (WASH), public health, and sustainable development in Africa. However, much of the available literature comprises observational studies, country-specific investigations, and reports from international organizations, with relatively limited longitudinal, implementation, and comparative research. In addition, because this was a narrative review rather than a systematic review, the literature selection and synthesis were not based on a formal risk-of-bias assessment, and relevant evidence may not have been captured despite the structured search strategy. Consequently, the findings should be interpreted as an integrated synthesis of the current evidence. Further high-quality longitudinal and comparative research is needed to strengthen context-specific recommendations and evaluate the long-term effectiveness of WASH interventions across diverse African settings.

Conclusion

WASH is a fundamental pillar of public health, health equity, and sustainable development in Africa. This review demonstrates that improving health outcomes requires moving beyond service coverage to ensure that WASH services are safe, reliable, equitable, climate-resilient, and sustainable. The synthesized evidence provides an important foundation for informing policy and practice by highlighting the need for integrated governance, sustained financing, infrastructure maintenance, community participation, and multisectoral collaboration to strengthen WASH systems across diverse African settings.

Despite the breadth of available evidence, important gaps remain regarding the long-term effectiveness, implementation, and scalability of WASH interventions in different African contexts. Future research should prioritize implementation and longitudinal studies that evaluate governance, financing, climate adaptation, and community-based strategies to generate context-specific

evidence to support equitable and sustainable WASH policies and programs.

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Author contributions

Abdullahi Hassan Elmi: Conceptualization, methodology, data curation, formal analysis, investigation, project administration, writing—original draft. Rasha Kadri Ibrahim: Methodology, supervision, writing, review, and editing, AHE and RKI: Investigation, data curation, writing—review and editing. All authors read and approved the final manuscript and agreed to be accountable for all aspects of the work.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participant

Not applicable for this study.

Consent for publication

Not applicable for this study.

Competing interests

The authors declare no competing interests related to this study.

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